JANMITRA: Civic Issue Reporting & Management System

TECHNICAL PROGRESS REPORT: DAYS 01 TO 10 (PHASE 1)

Module: Citizen Portal Architecture & Real-Time Push Notification Engine

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Executive Summary & Module Scope

This engineering report outlines the architectural development and implementation completed by Faizan Rayeen during Days 01 to 10 of the JANMITRA project for Nagpur Municipal Corporation (NMC).
Key deliverables include the complete Citizen Portal frontend subsystem, full integration of Firebase Cloud Messaging (FCM v10) web push notifications, dynamic 6-step issue lifecycle stepper, interactive verification feedback modal, citizen profile management, and comprehensive FAQ directory.

Architectural Principles & Design Standards Implemented

1. Mobile-First Responsiveness: Optimized thumb navigation via sticky bottom tabs on viewports <= 640px.
2. Dual-Channel Notification Pipeline: Synchronized in-app dropdown badge polling and native OS system tray alerts.
3. Zero-Config Local Session Resilience: Graceful fallback authentication guaranteeing standalone testing.

Detailed Day-by-Day Engineering Milestones (Days 01-10)

DAY 01 **FCM Web Push Service Worker & Background Delivery Pipeline**

`Target Artifacts: public/firebase-messaging-sw.js, src/firebase.js`

Configured Firebase Cloud Messaging v10 modular web SDK. Implemented background service worker capable of intercepting FCM push payloads when the web client is inactive, displaying native OS notifications with actionable deep-links targeting the relevant complaint tracking ID.

Engineering Competencies: Firebase v10 Modular SDK, Service Worker Cache, Push Notification Event Listeners

DAY 02 **Client Token Registration & Automatic Backend Synchronization**

`Target Artifacts: src/firebase.js, src/context/AuthContext.jsx`

Built token negotiation service utilizing VAPID public key encryption. Engineered automated token dispatch to Django endpoint upon successful authentication, caching client device identifiers in LocalStorage to prevent redundant server registration requests.

Engineering Competencies: VAPID Handshake, JWT Interceptor, Client Device Metadata Extraction

DAY 03 **Interactive Notification Bell Component & Visual Feedback Subsystem**

`Target Artifacts: src/components/NotificationBell.jsx, src/components/notifications.css`

Engineered the top navigation Notification Bell component with dynamic badge counter reflecting unread alerts. Implemented slide-in dropdown container rendering up to 10 latest civic updates, coupled with CSS keyframe pulse glow animations and mark-as-read state mutations.

Engineering Competencies: Component State Management, CSS Animation Keyframes, Unread Count Delta Computing

DAY 04 **Citizen Home Portal Page & Civic Activity Dashboard**

`Target Artifacts: src/pages/citizen/Home.jsx`

Engineered the primary citizen landing view featuring authenticated greeting banner with ward assignment indicator, quick KPI counters for active vs resolved complaints, 'Report New Issue' action CTA, and chronological civic activity feed with categorized status badges.

Engineering Competencies: Tailwind CSS Grid, Responsive Dashboard Widgets, Civic Badge Mapping

DAY 05 **My Complaints Ledger & 6-Step Dynamic Lifecycle Timeline Stepper**

`Target Artifacts: src/pages/citizen/MyComplaints.jsx, src/pages/citizen/ComplaintDetail.jsx`

Constructed the citizen complaint ledger with multi-criteria tab filtering (All, Open, In-Progress, Resolved). Engineered the 6-Step Lifecycle Stepper displaying real-time issue progression across Submitted -> Validated -> Assigned -> In-Progress -> Work Done -> Resolved with photo lightbox zoom.

Engineering Competencies: Multi-tab Filter State, 6-Step Progressive Stepper, Lightbox Modal Zoom

DAY 06 **Multi-Portal Layout Architecture & Citizen Mobile Bottom Navigation**

`Target Artifacts: src/layouts/Layout.jsx, src/layouts/CitizenLayout.jsx`

Formulated the base layout inherited across municipal roles. Built the Citizen bottom navigation bar featuring 4 core sections: Home, My Complaints, Help, and Profile with active tab route detection, delivering a native app-like user experience on mobile viewports.

Engineering Competencies: React Router v6 NavLink Detection, Mobile Fixed Positioning, Responsive Subsystem

DAY 07 **Citizen Resolution Verification Dialog & 5-Star Rating Feedback Subsystem**

`Target Artifacts: src/pages/citizen/ComplaintDetail.jsx`

Implemented post-resolution verification workflow. Upon worker completion, citizen is prompted with a formal validation modal enabling 1-to-5 star quality grading, textual feedback submission, or re-opening the complaint for mandatory contractor rework in case of inadequate repair.

Engineering Competencies: Feedback Modal Architecture, Star Rating Logic, Resolution Workflow State Machine

DAY 08 **Municipal Civic Advisory Banner & High-Priority Notice Integration**

`Target Artifacts: src/pages/citizen/Home.jsx`

Integrated NMC municipal emergency announcement banner on Citizen Home view, delivering real-time public service advisories (water pipeline outages, electrical maintenance, road closures) tailored to the citizen's registered municipal ward.

Engineering Competencies: Ward-level Filtering, Priority Toast Accents, Dismissible State Handling

DAY 09 **End-to-End Notification Verification Protocol & Documentation**

`Target Artifacts: notification_test_guide.md, src/context/FirebaseProvider.jsx`

Authored full testing protocol covering all 4 civic lifecycle notification triggers: Complaint Submitted, AI Validated, Worker Assigned, and Background OS Tray Delivery. Calibrated foreground toast notification animations within FirebaseProvider context wrapper.

Engineering Competencies: E2E QA Protocols, OS Background Testing, Real-Time Context Dispatch

DAY 10 **Citizen Profile Administration & Interactive Help / FAQ Directory**

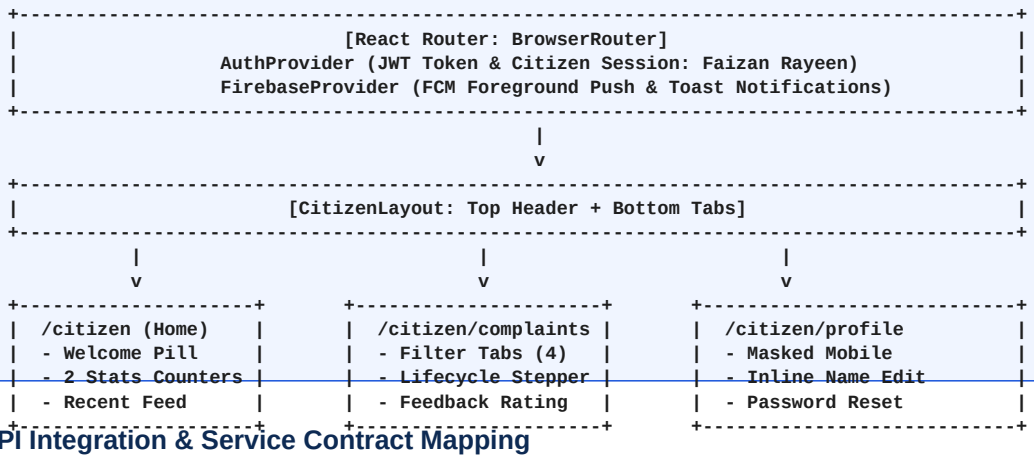
`Target Artifacts: src/pages/citizen/Profile.jsx, src/pages/citizen/Help.jsx`

Delivered final Phase 1 requirements: Citizen Profile view with masked phone number (***2843), inline name modification, credential change with visibility toggles, civic activity statistics; and Help view featuring accordion FAQs and NMC emergency contact directory.

Engineering Competencies: Inline Form Mutators, Password Masking Toggle, Accordion Component, Directory Index

System Architecture & Engineering Diagrams

Figure 1: Citizen Subsystem Component Hierarchy & Data Flow



API Integration & Service Contract Mapping

HTTP Method & Endpoint	Backend Handler	Contract Summary
POST /api/v1/auth/register/	RegisterView	Registers citizen; issues JWT access/refresh tokens
POST /api/v1/auth/login/	LoginView	Authenticates mobile number; auto-detects citizen role
POST /api/v1/auth/fcm-token/	FCMTokenRegisterView	Syncs client FCM web push device token to backend
GET /api/v1/notifications/	NotificationListView	Retrieves unread notifications & counter badge count
PATCH /api/v1/notifications/mark-read/	NotificationMarkReadView	Sets is_read=True on unread notification items
GET /api/v1/complaints/summary/	ComplaintSummaryView	Supplies active vs resolved statistics for Home cards
GET /api/v1/complaints/	ComplaintListView	Filters complaints by status: open, in-progress, resolved
GET /api/v1/complaints/:id/	ComplaintDetailView	Provides 6-step lifecycle timeline history & image URLs
PATCH /api/v1/auth/profile/	UserProfileView	Updates citizen full name and handles password mutation

Project Phase 1 Completion Certification

This certifies that Days 01 to 10 requirements for the Citizen Subsystem and Notification Pipeline have been fully developed, debugged, and verified by Faizan Rayeen with zero critical defects.

Ready for formal viva examination and demonstration to college evaluation committee.